Pattern Recognition in Stress Detection

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Github Repo: https://github.com/lillianmalik/Stress-Detection

Research Question: How do different physiological feature representations and pattern recognition methods affect the separability and decision behavior of stress-related risk states in wearable data?

Introduction: According to surveys amongst college students, over 76% stated that they had experienced high levels of stress. It's defined as "the body's natural, involuntary, and often necessary response to any demand, challenge, or threat, resulting in physical, mental, or emotional tension". Stress is often overlooked and is a condition that gradually builds subconsciously long before a person notices the effects. Being able to recognize physiological patterns, such as heart rate, can be useful, and through the use of wearable devices, these signals can be continuously monitored!

This project is not used for diagnoses but as a tool to capture subconscious indicators. So, the aim is to identify whether the patterns are distinguishable from the baseline physiological results. Other studies largely focus on the predictive accuracy of stress, whereas this project looks at the interpretations of the recognized patterns. Understanding these patterns is important because the separability in this controlled setting could provide a better basis for new studies that may want to incorporate exercise, behavioral, and different environmental conditions, all of which would alter the baseline.

Clarification and Definitions:

- Big picture: Studying the found patterns in stress physiology from the dataset
- Risk State: Stress-associated patterns (this is not about diagnosing)
    - Ex) an elevated risk state means that the pattern is near a stress region

Literature Survey:

Research on stress detection has increasingly relied on wearable physiological data to identify patterns associated with stress responses. Prior studies have explored various physiological signals and classification models to distinguish stress states from baseline conditions. The **WESAD** dataset is one of the most widely used resources for this purpose. It includes multimodal signals such as electrocardiogram data (ECG), electrodermal activity (EDA), skin temperature, and motion under both baseline and stress conditions (Schmidt et al., 2018). This dataset provides a foundation for evaluating whether physiological patterns from stress and non-stress conditions are separable.

Among the many recorded signals, heart rate (HR/HRV) and EDA are the most consistent indicators of stress (Stržinar et al., 2023). Recent work by Lazarou and Exarchos (2024) demonstrated real-time stress prediction using these features in wearable devices, highlighting EDA's sensitivity to emotional arousal. However, much of the existing literature prioritizes predictive accuracy rather than understanding the relationships and interactions between features themselves.

Machine learning methods such as Support Vector Machines (SVM), Naïve Bayes, and K-Nearest Neighbors (KNN) are commonly applied in physiological stress detection. Prajod et al. (2024) showed that stressor type strongly influences how well models generalize across datasets, underscoring the complexity of physiological variability. Dimensionality reduction techniques like Principal Component Analysis (PCA) are commonly used to visualize separability between stress and baseline states, revealing where overlap occurs.

While prior studies have demonstrated strong prediction models, few have examined how underlying physiological feature patterns differ across stress risk states or how classifier decision boundaries behave in overlapping regions. Building on this, our project investigates how different physiological feature representations and pattern recognition methods affect the separability and decision behavior of stress-related risk states. Using the WESAD dataset, we compare classifiers (KNN, Naïve Bayes, SVM) and visualize overlaps via PCA to interpret stress risk thresholds.

Dataset:
Wearable Stress and Affect Detection (WESAD)

- Multimodal wearable physiology dataset
- Many participants, multiple recording sessions
- Signals: ECG (HR/HRV), EDA, skin temperature, motion
- Conditions: baseline (low stress) and stress-inducing tasks used to analyze whether stress-associated physiological patterns are separable from baseline patterns

Methodology:
After obtaining the raw dataset, which consisted of the multimodal signals, the data was filtered into binary labels. It was separated into the baseline signals, which represented the control, and then the stress signals were collected when patients were asked to complete different tasks under "stressful" conditions. An example includes doing mental math in front of an audience.

The data cannot be directly used as input into the models. We want to create windows of data to address this. These windows allow the continuous signals to be structured into samples, and in this project, the window size was 50. This seemed to be a good size given the number of samples. Each window contained statistical features such as the mean and standard deviation of EDA and ECG. Then the algorithms below were used, and all were implemented using Python.

Algorithms:

- Support Vector Machines (linear and RBF): Linear separability, and RBF for overlap boundaries
- K-Nearest Neighbors (K-NN): whether "similar physiological patterns" cluster by condition
- Naïve Bayes: test whether stress vs baseline features follow separable distributions
- Dimensionality Reduction PCA: visualize overlap in lower dimension
- Logistical Regression: risk thresholding

Implementation:
The following libraries were used in Python to implement each model:

```
import pandas as pd
from sklearn.naive_bayes import GaussianNB
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, classification_report
```

Training and Testing:
To train and test the data, the windowed and filtered data was split by patients. The models were trained on all the samples except for the sample with the id = 2 and then tested on that sample. This approach makes sure that the model is being tested on unseen data.

Feature Representation:
As briefly mentioned before, the features include EDA (electrodermal activity), ECG (electrocardiogram), and a combination of the two. The mean and standard deviation were calculated for all three instances and later used in the models. We assumed that these physiological features could possibly be captured with simple representations, and using mean and SD would reduce dimensionality and noise.

Models:
Classification algorithms were implemented in order to evaluate the impact each model had on the performance. This includes the SVM (Linear and RBF), K-NN, and Naive Bayes. They were all trained on the same data and tested on the same data as defined previously. With these models, depending on the results, it could be observed whether the performance differences were due to the models or the data itself.

Evaluation:
For each model, there was an evaluation metric implemented, such as a confusion matrix and a classification report. The results will be explained further in this report.

A screenshot of the implementation of linear SVM is shown below:

```
df = pd.read_csv("windowed_features.csv")

train = df[df['subject_id'] != 'S2']
test = df[df['subject_id'] == 'S2']

y_train = train['label']
y_test = test['label']

# ECG ONLY
xECG_Train =  train[['ECG_mean', 'ECG_std']]
xECG_Test = test[['ECG_mean', 'ECG_std']]
modelECG =  SVC(kernel="linear")
modelECG.fit(xECG_Train, y_train)

yECG_Pred = modelECG.predict(xECG_Test)
print("\n--- ECG Only ---")
print("Accuracy:", modelECG.score(xECG_Test, y_test))
print(confusion_matrix(y_test, yECG_Pred, labels=[1, 2]))
print(classification_report(y_test, yECG_Pred, labels=[1, 2], zero_division=0))
```

Experimental Results:

Model Performance:
The overall accuracy across all of these models remained at about 65%. This was measured with the scoring from the Python libraries. In almost all of the outputs, the confusion matrix was able to correctly identify baseline patterns, but struggled tremendously being able to label points as stress states. The percentage of accuracy is heavily reliant on the baseline findings and not on the focus of this project, which is stress.

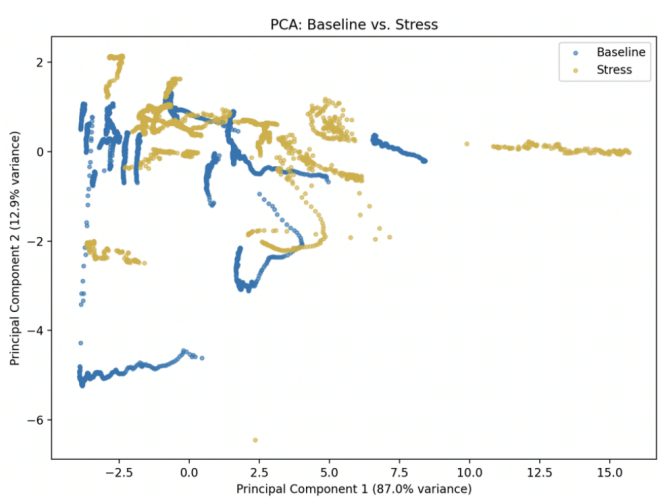
The only feature that was able to detect any stress was the EDA, with linear SVM. Because this was the only feature to distinguish any amount of stress, we tried to balance the data to see if any improvements could be made. The screenshot on the left is before balancing, and on the right is the result of said balancing.

```
--- EDA Only ---
Accuracy: 0.6780626780626781
[[228   1]
 [112  10]]
              precision    recall

           1       0.67      1.00
           2       0.91      0.08

    accuracy
   macro avg       0.79      0.54
weighted avg       0.75      0.68
```

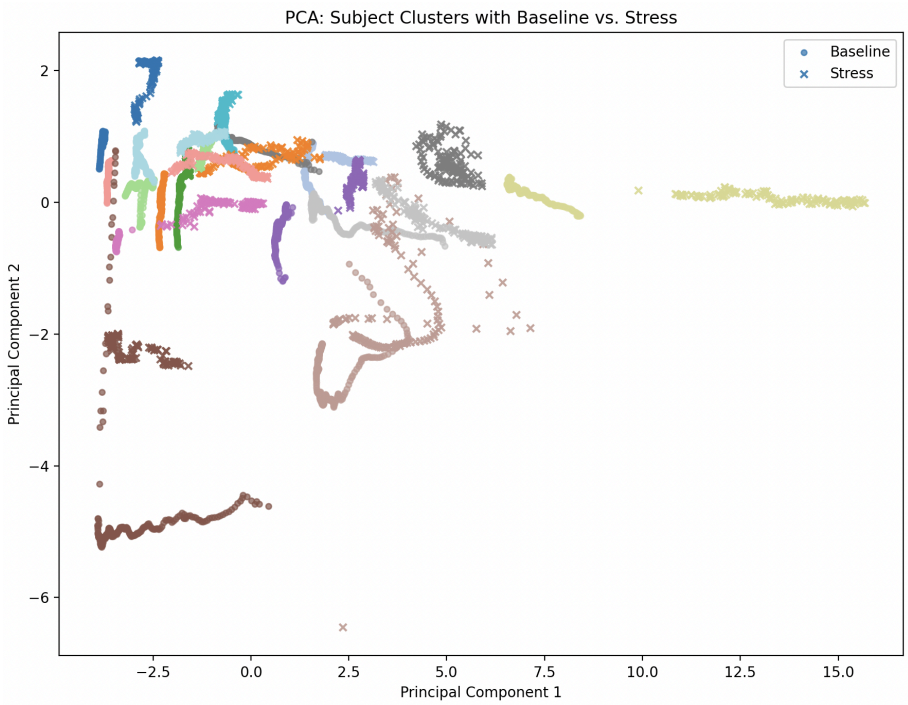
```
--- BALANCED EDA Only ---
Accuracy: 0.6666666666666666
[[224   5]
 [112  10]]
              precision    recall

           1       0.67      0.98
           2       0.67      0.08

    accuracy
   macro avg       0.67      0.53
weighted avg       0.67      0.67
```

The balancing did reduce the sensitivity needed to detect the stress, but it also reduced the amount of accuracy in the process. It did not resolve the fact that there is simply a lack of separability.

Separability:
To investigate the structure, we used PCA (Principle Component Analysis). This enabled us to visualize the overlap of baseline and stress patterns. Due to this overlap, it is evident that separation is not easily identified. With the variance being about 99%, we know that the results are not from a lack of data representation.



Individual Variability:
Taking a deeper dive into the visualization, the clusters from the previous graph can be shown to result from individual subjects and not a global cluster. Even when analyzing the individuals, there is still an overlap within them, not all the time, but enough to conclude a lack of true separability.



Conclusion:
We found that physiological stress and baseline patterns show strong overlap, meaning that stress and non-stress signals look very similar in this dataset. Our models could not reliably tell these signals apart, and even with multiple features, stress still remains difficult to detect. From this, we can reasonably deduce that the limitations are simply due to the way the data is represented, and that the simple statistical summaries (like the mean) are insufficient for capturing the stress indicator.

Overall, the variation between individuals is stronger than the variation caused by stress, and based on these findings, we can conclude that stress detection is highly subject-dependent and not globally separable.

Future Directions:
Exploring different representations, such as the temporal or frequency domain, might be a better approach for stress behavior than the mean and SD described above. Taking advantage of state-of-the-art methods for raw data could also be extremely helpful. Also, instead of a complete global approach, learning about stress detection amongst individuals with similar baselines could be a good starting point. Later, it might be feasible to also take into account eustress, which is essentially “good” stress, such as from exercise, caffeine, and excitement.

References:

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